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Software Requirements Specification for UniQueue

by:

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May 2026

1. Project Charter

Project Title

UniQueue: An Online Queue and Appointment Management System for School Services

Project Objective

To develop a web-based system that reduces overcrowding and long waiting times in school offices by allowing students to schedule appointments and monitor queue status online.

SMART Success Criteria

- Reduce average waiting time by at least 50%
- Support at least 200 concurrent users
- Provide real-time queue updates within 10 seconds
- Minimize walk-in congestion during office hours

High-Level Risks

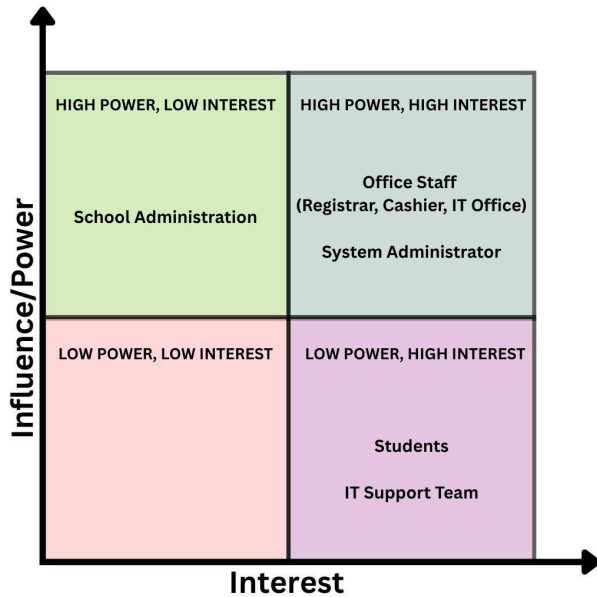
- Internet connectivity issues
- Resistance from users adapting to digital systems
- Limited development time and resources
- Incorrect queue updates by staff

2. Stakeholder Analysis

Interest/Influence Matrix

Stakeholder	Type	Responsibility / Interest	Influence (H/M/L)
Students	Primary User	Submit requests, monitor queue, attend appointments	Medium
Office Staff (Registrar, Cashier, IT)	Primary User	Manage queues, process requests, update status	High
System Administrator	Technical/Admin	Manage accounts, configure system settings, monitor reports	High
School Administration	Decision-Maker	Approve system implementation, allocate budget	High

IT Support Team	Support	Maintain system, troubleshoot issues	Medium



Negotiation Log

Concern	Stakeholders	Resolution
Queue overflowing	Students	Appointment scheduling
Staff workload	Office staff	Automated queue monitoring
Data privacy	Administration	Role based control

3. Requirements Elicitation

Interview Protocols

Interview Participants

- Registrar Staff
- Cashier Staff
- Students

Sample Questions

1. What problems do you experience in manual queuing?
2. How are appointments currently managed?
3. What causes delays in processing students?
4. How do students receive updates regarding their requests?

Summary

- students experience long waiting times.
- Walk-ins create overcrowding in offices.
- Staff struggle with manual queue management.
- Communication through email is often delayed.
- Students prefer a digital system with real-time updates and appointment scheduling.

4. Specific Requirements

Functional Requirements (FR)

ID	Requirements	Priority
FR-1	The system shall allow users to securely log in using valid credentials.	Must Have
FR-2	The system shall allow students to book appointments online.	Must Have
FR-3	The system shall display real-time queue status updates.	Must Have
FR-4	The system shall allow office staff to manage and update queues.	Must Have
FR-5	The system shall allow students to reschedule appointments.	Should Have
FR-6	The system shall send appointment and queue notifications to users.	Should Have
FR-7	The system shall generate reports for queue transactions and appointments.	Could Have
FR-8	The system shall provide analytics for office performance monitoring.	Could Have
FR-9	The system shall support online payment integration for school services .	Won't Have (Current Version)

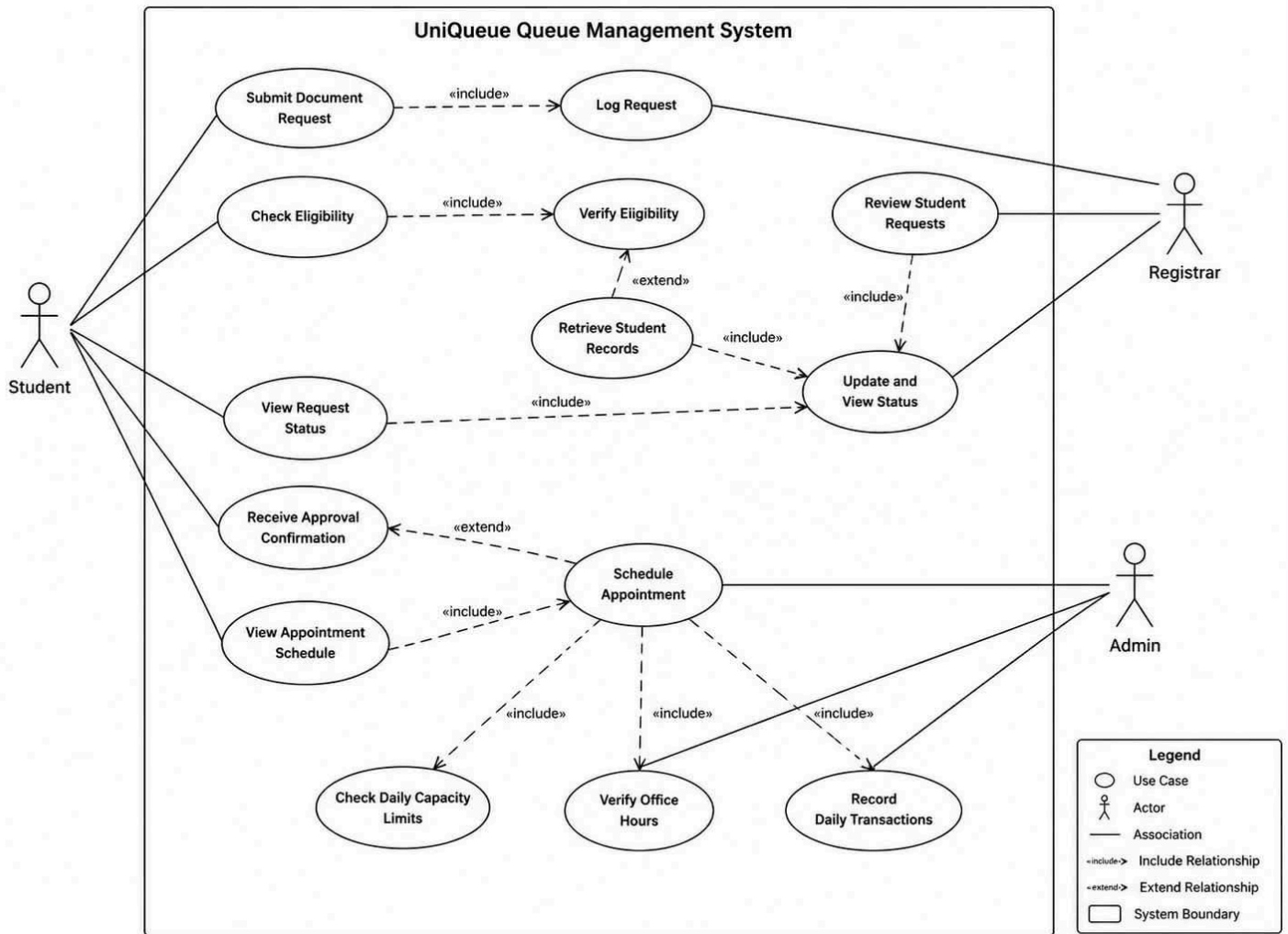
Non-Functional Requirements (NFR)

ID	Requirements	Category
NFR-1	The system shall respond to user requests within 5 seconds.	Performance
NFR-2	The system shall support at least 200 concurrent users.	Scalability

NFR-3	The system shall use HTTPS/TLS encryption for secure communication.	Security
NFR-4	The system shall be accessible on desktop and mobile devices.	Usability
NFR-5	The system shall maintain 99% system availability during office hours.	Reliability
NFR-6	The system shall automatically back up queue and appointment data daily.	Backup & Recovery
NFR-7	The system shall provide role-based access control for administrators and staff.	Security
NFR-8	The system shall have a user-friendly interface that is easy to navigate.	Usability

5. Logical Use Case Models

Use Case Diagram



Detailed Narrative Descriptions for Must-Have Features

Use Case 1: Secure Login

Item	Description
Primary Actor	Student / Office Staff / Administrator
Description	Users log into the system securely using their account credentials.
Preconditions	User accounts already exist.
Trigger	User enters username and password.
Main Flow	1. The user opens the login page. 2. User enters username and password. 3. System validates credentials. 4. The system grants access to the dashboard.
Alternative Flow	If credentials are incorrect, the system displays an error message.

Postconditions	The user successfully accesses the system.
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Use Case 2: Book Appointment

Item	Description
Primary Actor	Student
Description	Students schedule an appointment for a school service.
Preconditions	Students are logged into the system.
Trigger	Student selects "Book Appointment."
Main Flow	1. The student selects office/service. 2. Students choose available dates and time. 3. Student submits appointment request. 4. The system saves appointment details.
Alternative Flow	If the schedule is unavailable, the system asks students to choose another slot.
Postconditions	Appointment is successfully recorded in the system.

Use Case 3: View Real-Time Queue Status

Item	Description
Primary Actor	Student
Description	Students monitor current queue updates online.
Preconditions	Students are logged into the system.
Trigger	Student opens queue monitoring page.
Main Flow	1. The system displays the current queue number. 2. System updates queue status automatically. 3. Students monitor waiting progress.
Alternative Flow	If internet connection is lost, queue updates may be delayed.
Postconditions	Students receives updated queue information.

Use Case 4: Manage Queue

Item	Description
Primary Actor	Office Staff

Description	Office staff update and manage student queue records.
Preconditions	Staff is logged into the system.
Trigger	The staff opens a queue management panel.
Main Flow	1. Staff views the active queue list. 2. The staff calls the next student. 3. Staff updates queue status. 4. The system refreshes queue information in real time.
Alternative Flow	If incorrect data is entered, the system requests correction.
Postconditions	Queue records are updated successfully.

Use Case 5: Manage User Accounts

Item	Description
Primary Actor	System Administrator
Description	Administrator manages system accounts and access permissions.
Preconditions	Administrator is logged in.
Trigger	Administrator selects account management module.
Main Flow	1. Admin views user list. 2. Admin creates, updates, or deletes accounts. 3. The system saves account changes.
Alternative Flow	If a duplicate account exists, the system displays a warning.
Postconditions	User accounts are updated successfully.

6. System Boundary

Context Diagram / DFD

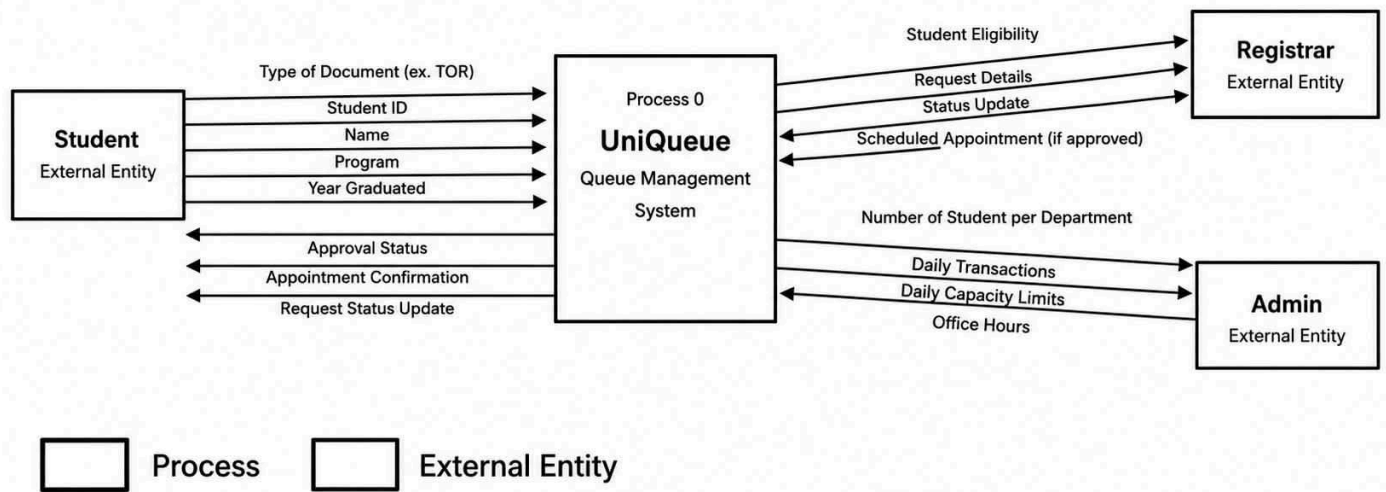


Figure 1: Level-0 Context DFD — UniQueue Queue Management System

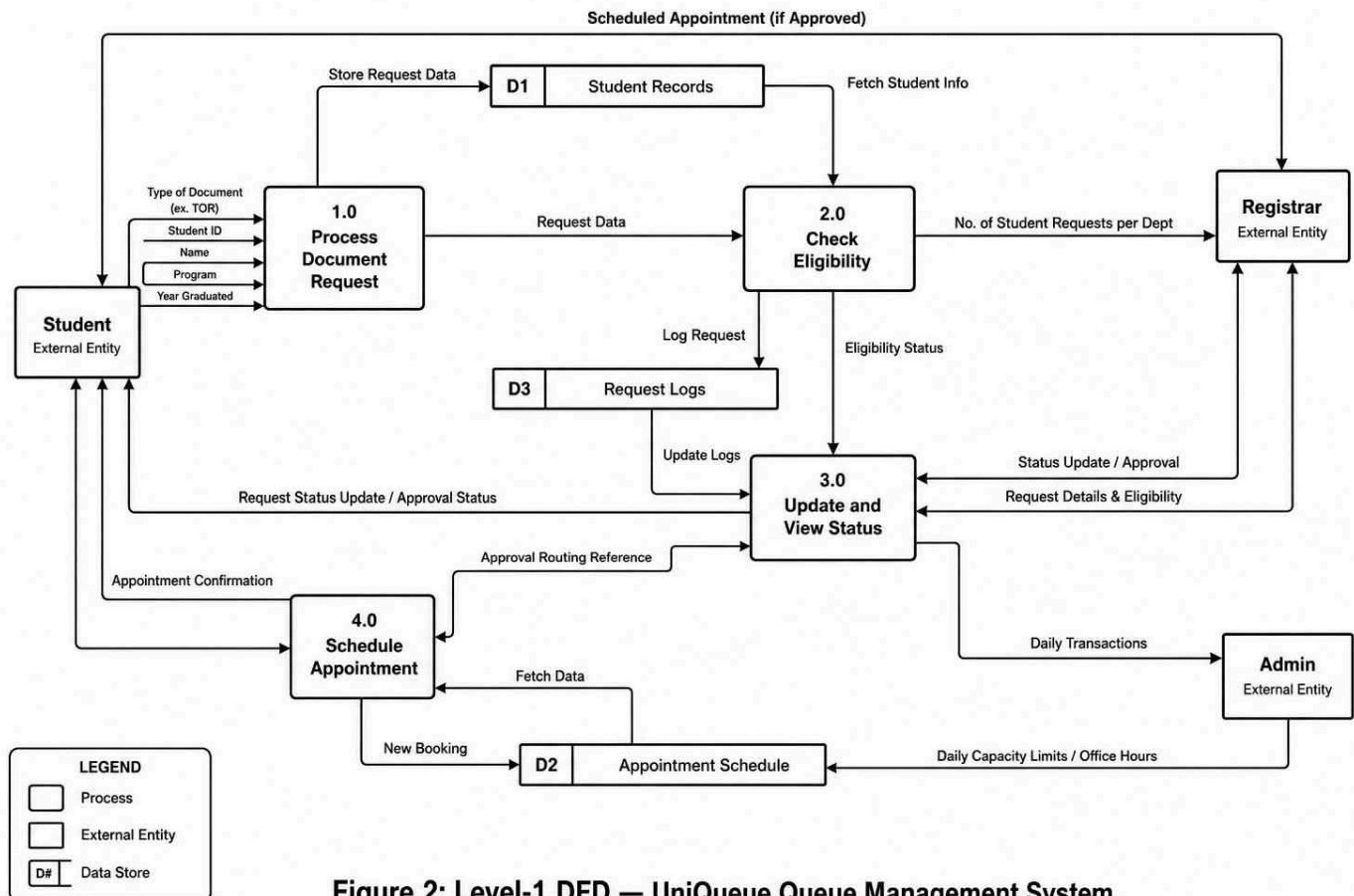


Figure 2: Level-1 DFD — UniQueue Queue Management System

